SmartCold Guard: ESP32-Based IoT Cold Chain Monitoring and Security System

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1. Project Overview

Project Title

SmartCold Guard: ESP32-Based IoT Cold Chain Monitoring and Security System

Project Summary

SmartCold Guard is an IoT-enabled cold-chain monitoring and security system designed to improve the safety, reliability, and transparency of transporting temperature-sensitive products such as food items, medicines, and biological materials.

The system uses an **ESP32 microcontroller** as the main processing unit, connected with environmental, location, and security sensors. The collected data is transmitted to a **custom-built IoT dashboard** where users can remotely monitor storage conditions, track location, review access events, and receive alerts.

Unlike existing solutions that depend on third-party platforms, this project develops an independent dashboard platform that provides better customization, scalability, and control over collected data.

The system aims to provide a low-cost alternative to commercial cold-chain monitoring systems while maintaining essential features such as real-time monitoring, theft prevention, and remote accessibility.

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2. Functional Requirements

The system should provide the following functionalities:

2.1 Environmental Monitoring

Temperature Monitoring

- Continuously measure storage temperature using a DHT11 temperature and humidity sensor.
- Send real-time temperature data to the dashboard.

- Trigger alerts when temperature exceeds predefined limits.

Humidity Monitoring

- Monitor humidity levels inside the storage container.
- Display current and historical humidity data.
- Detect unsuitable storage conditions.

2.2 Real-Time Location Tracking

GPS Tracking

- Collect live location coordinates using the NEO-6M GPS module.
- Send latitude, longitude, and timestamp information to the dashboard.
- Allow users to track transportation routes.

Location History

- Store previous location records.
- Display movement history of the storage unit.

2.3 Secure Access Control

RFID Authentication

- Implement RFID-based access control using the MFRC522 module.
- Allow only authorized users to access the storage container.
- Record access attempts.

Access Management

- Store authorized RFID card information.
- Display access logs on the dashboard.

2.4 Theft and Tamper Detection

Unauthorized Opening Detection

- Use an LDR sensor to detect sudden light exposure caused by opening the storage box.

- Generate security alerts during unauthorized access.

Object Presence Detection

- Use an ultrasonic sensor to monitor container status and object presence.

Alarm System

- Activate a buzzer during security violations.
- Send notification alerts to the dashboard.

2.5 Custom IoT Dashboard

The project will develop a web-based monitoring dashboard instead of using Blynk.

The dashboard will provide:

- Real-time sensor visualization.
- Temperature and humidity graphs.
- GPS location tracking.
- Security status monitoring.
- RFID access history.
- Alert notifications.
- Device connectivity status.

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3. Non-Functional Requirements

3.1 Performance

- The system should collect sensor data continuously.
- Data transmission should occur with minimum delay.
- Dashboard updates should happen in near real-time.

3.2 Reliability

- The system should operate continuously during transportation.
- Sensor failures should be detected.

- Data loss should be minimized during temporary network failures.

3.3 Security

- User authentication should be implemented for dashboard access.
- Communication between ESP32 and server should be protected.
- Unauthorized users should not access monitoring data.

3.4 Scalability

The system architecture should support future expansion:

- Multiple cold storage units.
- Additional sensors.
- More user accounts.
- Cloud-based data storage.

3.5 Usability

The dashboard should provide:

- Simple user interface.
- Clear visualization.
- Easy access to alerts and reports.
- Mobile-friendly design.

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4. MVP (Minimum Viable Product) Definition

The minimum working version of SmartCold Guard will include:

Hardware MVP

- ESP32 microcontroller.
- DHT11 temperature and humidity sensor.
- RFID access module.

- GPS module.
- LDR theft detection sensor.
- Buzzer alarm.

Software MVP

A custom dashboard capable of:

- Receiving sensor data from ESP32.
- Displaying live temperature and humidity.
- Showing GPS location.
- Displaying security alerts.
- Showing RFID access events.

MVP Demonstration

The prototype will demonstrate:

1. Temperature monitoring inside a storage box.
2. Real-time data transmission.
3. Unauthorized opening detection.
4. RFID-based access control.
5. Dashboard-based remote monitoring.

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5. System Architecture

5.1 Major Components

ESP32 Microcontroller

Purpose:

- Collect sensor data.
- Process input signals.
- Communicate with server.
- Control alarm systems.

Advantages:

- Built-in WiFi and Bluetooth.
- Low power consumption.
- Affordable IoT development platform.

Sensors

DHT11 Sensor

Function:

- Temperature measurement.
- Humidity measurement.

NEO-6M GPS Module

Function:

- Real-time location tracking.
- Transportation route monitoring.

MFRC522 RFID Module

Function:

- User authentication.
- Access control.

LDR Sensor

Function:

- Detect unauthorized opening.

HC-SR04 Ultrasonic Sensor

Function:

- Detect object presence.
- Monitor container status.

5.2 Data Flow

1. Sensors collect environmental and security information.
2. ESP32 processes collected sensor values.
3. ESP32 sends data through WiFi.
4. Backend API receives and stores data.
5. Database maintains historical records.
6. Dashboard displays live information.
7. Alert system notifies users when abnormal events occur.

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6. Technology Selection

Hardware Technology Selection

Option	Advantages	Limitations	Decision
Arduino Uno	Cheap and simple	No built-in WiFi	Not selected
Raspberry Pi	High processing capability	Expensive and higher power consumption	Not selected
ESP32	Built-in WiFi, low cost, IoT optimized	Limited processing power	Selected

Decision:
ESP32 is selected because it provides sufficient processing capability and wireless communication for IoT applications at low cost.

Software Technology Selection

Dashboard Framework

Option	Advantages	Limitations	Decision
Blynk	Fast development	Limited customization and dependency on external platform	Not selected

Option	Advantages	Limitations	Decision
Node.js + Web Dashboard	Flexible, scalable, customizable	Requires backend development	Selected
Mobile-only Application	Good portability	More development complexity	Not selected

Decision:
A custom web dashboard using a backend API will be developed to provide full control over data visualization and system expansion.

6.1 Proposed Software Stack

Frontend Dashboard

Possible technologies:

- React.js
- HTML/CSS/JavaScript
- Chart libraries for visualization

Functions:

- Live sensor display.
- Graph visualization.
- Map integration.
- Alert interface.

Backend Server

Possible technologies:

- Node.js
- Express.js

Responsibilities:

- Receive ESP32 data.
- Provide API services.
- Handle user authentication.

- Manage communication between device and dashboard.

Database

Possible options:

- MongoDB
- MySQL
- Firebase Database

Purpose:

- Store sensor history.
- Maintain user records.
- Store security events.

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7. Project Risks and Solutions

Risk	Impact	Solution
Network failure	Data transmission interruption	Add local data storage and reconnect mechanism
Sensor accuracy issues	Incorrect monitoring results	Sensor calibration
GPS signal loss	Location unavailable	Store previous location data
Power consumption	Short operating time	Battery optimization
Security vulnerabilities	Unauthorized dashboard access	Authentication and encryption

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8. Future Improvements

Future development can include:

Advanced Sensors

- MPU6050 for shock and orientation detection.

- Higher accuracy temperature sensors.

Communication Improvements

- SIM800L GSM module for communication without WiFi.
- LoRa communication for long-distance monitoring.

Data Analytics

- AI-based temperature prediction.
- Automatic failure detection.
- Predictive maintenance.

Hardware Improvements

- Rechargeable battery system.
- Weather-resistant enclosure.
- Industrial-grade sensors.

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9. Expected Outcome

At the completion of this project, the team expects to deliver:

- A functional IoT-based cold-chain monitoring prototype.
- A custom web dashboard for real-time monitoring.
- Secure RFID-based access control.
- Theft and tamper detection system.
- GPS-based tracking capability.
- A scalable platform for future industrial applications.